

# PROJECT REPORT

**Project Title: Import Data Using Transform Maps in ServiceNow**

## 1. INTRODUCTION

### 1.1 Project Overview

This project demonstrates an end-to-end employee data import process in ServiceNow using Import Sets and Transform Maps. Employee information is prepared in an Excel/Google Spreadsheet, loaded into an Import Set table, mapped to a custom Employee Test table, and transformed into usable ServiceNow records. The project also includes Coalesce configuration to prevent duplicate records and Reports and Dashboards to provide a centralized view of employee information and import activity.

### 1.2 Purpose

The purpose of the project is to understand and implement bulk data migration in ServiceNow accurately and efficiently. It provides practical knowledge of loading external spreadsheet data, creating a staging table, mapping source and target fields, executing transformations, updating existing records, avoiding duplicates, and presenting the imported information through reports and dashboards.

## 2. IDEATION PHASE

### 2.1 Problem Statement

Organizations may receive employee information in spreadsheet format and need to move large amounts of data into ServiceNow. Manual entry is time-consuming and can introduce errors or duplicate records. A structured import process is therefore required to load external data, map it correctly, update existing records, and maintain reliable employee information.

### 2.2 Empathy Map Canvas

| Think & Feel                                                                 | See                                                                    | Hear                                                                      | Do                                                                                         |
|------------------------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Needs accurate employee records and a simple way to manage repeated imports. | Excel files, ServiceNow tables, import status, reports and dashboards. | Requirements for accurate migration, duplicate prevention and validation. | Prepares spreadsheet data, loads it, maps fields, transforms records and verifies results. |

### 2.3 Brainstorming

The project idea was developed around a simple automated workflow: prepare structured employee data, create a custom target table, load the spreadsheet into an Import Set, create a Transform Map, map source fields to target fields, execute the transformation, use Coalesce for duplicate prevention, and create reports and a dashboard for monitoring.

## 3. REQUIREMENT ANALYSIS

### 3.1 Customer Journey map

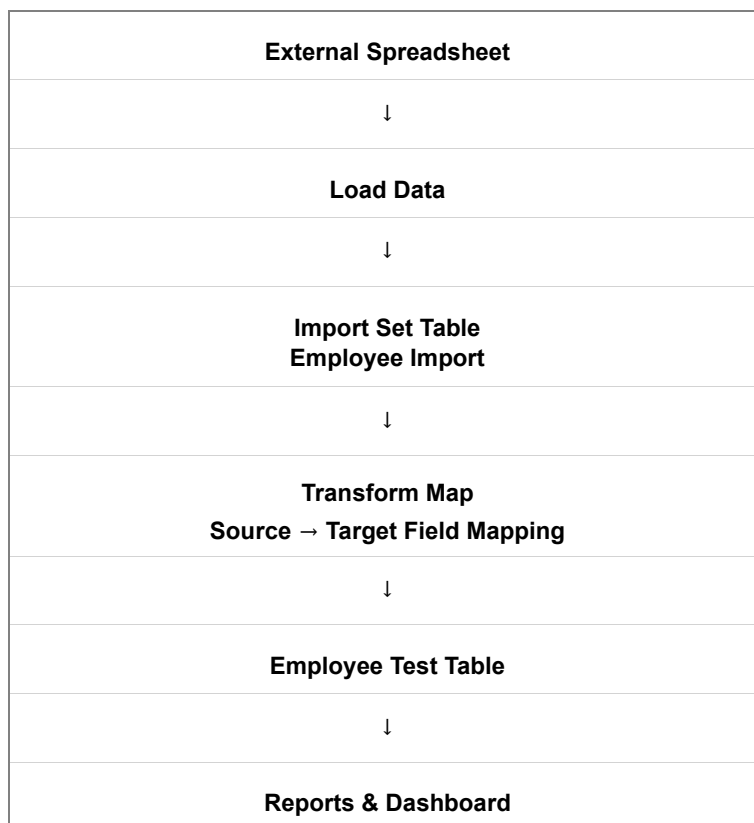
| Stage        | User Activity                                   | System Response                               |
|--------------|-------------------------------------------------|-----------------------------------------------|
| Prepare Data | Create spreadsheet with employee details.       | Spreadsheet is ready for import.              |
| Load Data    | Use Load Data and select the Excel file.        | Data is loaded into the Import Set table.     |
| Map Data     | Configure the Transform Map and field mappings. | Source fields are connected to target fields. |

| Stage     | User Activity                                    | System Response                             |
|-----------|--------------------------------------------------|---------------------------------------------|
| Transform | Run the transformation.                          | Records are inserted or updated.            |
| Validate  | Check transform history and Employee Test table. | Import status and records can be verified.  |
| Analyze   | Open reports and dashboard.                      | Employee information is presented visually. |

### 3.2 Solution Requirement

- ServiceNow instance with access to tables, Import Sets, Transform Maps, Reports and Dashboards.
- Excel/Google Spreadsheet containing structured employee data.
- Custom target table: Employee Test (u\_employee\_test).
- Import Set table: Employee Import (u\_employee\_import).
- Transform Map connecting Employee Import to Employee Test.
- Required fields: Employee ID, Employee Name, Email, Department and Location.
- Coalesce enabled on a suitable unique field such as Employee ID.

### 3.3 Data Flow Diagram



### 3.4 Technology Stack

| Technology / Tool | Purpose                                                          |
|-------------------|------------------------------------------------------------------|
| ServiceNow        | Platform for data import, transformation, storage and reporting. |
| Import Sets       | Staging area for external spreadsheet data.                      |
| Transform Maps    | Maps source fields to target fields and transforms records.      |

|                            |                                                            |
|----------------------------|------------------------------------------------------------|
| Excel / Google Spreadsheet | Source of structured employee data.                        |
| Reports                    | Visualizes employee data by department, location and list. |
| Dashboard                  | Combines reports into a centralized view.                  |

## 4. PROJECT DESIGN

### 4.1 Problem Solution Fit

The project directly addresses spreadsheet-to-ServiceNow data migration. Instead of entering employee records manually, the Import Set provides a staging area and the Transform Map automates field mapping and record creation or updating. Coalesce further supports repeated imports by identifying existing records and avoiding unnecessary duplicates.

### 4.2 Proposed Solution

- Create the Employee Test custom table with the required employee fields.
- Create the Employee Import Import Set table.
- Load employee spreadsheet data into the Import Set.
- Create the Sample Spreadsheet Import Transform Map.
- Map Employee ID, Employee Name, Email, Department and Location.
- Run the transform and verify the imported records.
- Enable Coalesce to update matching employee records instead of inserting duplicates.
- Create department, location and employee list reports and combine them in a dashboard.

### 4.3 Solution Architecture

**Data Source Layer:** Excel/Google Spreadsheet → **Staging Layer:** Employee Import Import Set → **Transformation Layer:** Transform Map + Field Maps + Coalesce → **Target Layer:** Employee Test Table → **Presentation Layer:** Reports + Employee Analytics Dashboard.

## 5. PROJECT PLANNING & SCHEDULING

### 5.1 Project Planning

| Phase   | Planned Activity                                                |
|---------|-----------------------------------------------------------------|
| Phase 1 | Prepare spreadsheet data.                                       |
| Phase 2 | Create Employee Test custom table and fields.                   |
| Phase 3 | Create and load the Employee Import table.                      |
| Phase 4 | Create Transform Map and configure field mappings.              |
| Phase 5 | Transform data and validate records.                            |
| Phase 6 | Enable Coalesce and test duplicate prevention.                  |
| Phase 7 | Import additional Excel data and verify insert/update behavior. |
| Phase 8 | Create reports.                                                 |
| Phase 9 | Create dashboard and add reports.                               |

## 6. FUNCTIONAL AND PERFORMANCE TESTING

### 6.1 Performance Testing

- Verify that spreadsheet data is successfully loaded into the Import Set table.

- Verify that all required source and target fields are mapped correctly.
- Verify that the Transform operation completes successfully.
- Verify that imported records appear in the Employee Test table.
- Verify that Coalesce updates an existing employee record when the matching Employee ID is imported again.
- Verify that repeated import of the same data does not create additional duplicate records.
- Verify that reports display department-wise, location-wise and employee-list information.
- Verify that all three reports are successfully displayed on the Employee Analytics Dashboard.

**Observed validation from the guideline workflow:** during the repeated-data test, the transform history can show inserted and updated records for changed/new data, while re-importing the same unchanged sheet can result in ignored records. This demonstrates the intended duplicate-control behavior.

## 7. RESULTS

### 7.1 Output Screenshots

The screenshot displays the ServiceNow interface for the 'Table - Employee Test'. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', and 'Admin'. The main header shows 'Table - Employee Test' with a search bar and buttons for 'Delete', 'Update', and 'Delete All Records'. Below the header, there is a section for 'Table Columns' with a search bar and a 'New' button. The 'Table Columns' section contains a table with the following data:

| Column label  | Type          | Reference | Max length | Default value | Display |
|---------------|---------------|-----------|------------|---------------|---------|
| Email         | String        | (empty)   | 40         | false         |         |
| Updated by    | String        | (empty)   | 40         | false         |         |
| Updates       | Integer       | (empty)   | 40         | false         |         |
| Employee ID   | String        | (empty)   | 40         | false         |         |
| Updated       | Date/Time     | (empty)   | 40         | false         |         |
| Location      | String        | (empty)   | 40         | false         |         |
| Sys ID        | Sys ID (GUID) | (empty)   | 32         | false         |         |
| Department    | String        | (empty)   | 40         | false         |         |
| Created by    | String        | (empty)   | 40         | false         |         |
| Employee Name | String        | (empty)   | 40         | false         |         |
| Created       | Date/Time     | (empty)   | 40         | false         |         |

Below the table, there are buttons for 'Delete', 'Update', and 'Delete All Records'. The 'Related Links' section includes links to 'Form Builder', 'Design Form', 'Layout Form', 'Layout List', 'Show Form', 'Show List', 'Show Schema Page', 'Add to Service Catalog', 'Run Point Scan', and 'Explore REST API'. The bottom navigation bar shows 'Access Controls (4)', 'Security Data Filters', 'Labels (1)', 'Database Indexes (1)', and 'Table Subscription Configuration (1)'. The bottom right corner has a search bar and a button for 'Actions on selected rows...'.

←

https://dev402677.service-now.com/now/nav/ui/classic/params/target/import\_status.do%3Fsysparm\_ds\_sysid%3D856c864b93270f10288b5f08bba10d%26sysparm\_create\_transform\_map%3Dtrue%26sysparm\_table%3Du\_employee...

Click to go back (Alt+Left arrow), hold to see history

WorkspacesAdmin

ServiceNow

Search

Verify it's you

Chat

Progress

Name

ImportProcessor

State

Complete

Completion code

Success

Message

Processed 15, inserts 15, updates 0, errors 0, empty and ignored 0, ignored errors 0 (0:00:01.336)

Next steps...

Import sets

Go to the import sets for this data load

Loaded data

Go to the newly imported data inside the staging table: u\_employee\_import

Create transform map

Create a transform map for the newly staged data

Import log

View the import log

servicenow

AllFavoritesHistoryWorkspacesAdmin

ServiceNow

Search

Progress

Name

Transforming: ISET0010006

State

Complete

Completion code

Success

Message

Transformation complete

Next steps...

ISET0010006

Go to the import sets for this data load

Transform history

Show the transform history, related errors and log

Import log

View the import log

servicenow

AllFavoritesHistoryWorkspacesAdmin

Import Sets

Search

Search

Import Sets

Number

Search

Actions on selected rows...

All

|  | Number       | Short description         | State     | Import set table                               | Data source                              | Created             | Load completed      | Load run time |
|--|--------------|---------------------------|-----------|------------------------------------------------|------------------------------------------|---------------------|---------------------|---------------|
|  | ISSET0010001 | Type: File<br>Format: CSV | Processed | Employee Bulk Import [u_employee_test_staging] | Employee Bulk Import                     | 2026-09-28 04:15:19 | 2026-09-28 04:15:19 | 0 Seconds     |
|  | ISSET0010002 | Type: File<br>Format: CSV | Processed | Employee Bulk Import [u_employee_test_staging] | banking customer records spreadsheet.csv | 2026-09-28 07:07:17 | 2026-09-28 07:07:17 | 0 Seconds     |
|  | ISSET0010003 | Type: File<br>Format: CSV | Processed | Employee Bulk Import [u_employee_test_staging] | banking customer records spreadsheet.csv | 2026-09-28 07:23:24 | 2026-09-28 07:23:24 | 0 Seconds     |

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servicenow

AllFavoritesHistoryWorkspacesAdmin

Table Transform Map - Employee Staging to Employee Test Map

Search

Search

Table Transform Map - Employee Staging to Employee Test Map

CopyUpdateDelete

\* NameEmployee Staging to Employee Test Map

\* Source tableEmployee Bulk Import [u\_employee\_test\_st...

Active

Run business rules

Enforce mandatory fieldsNo

Copy empty fields

Create new record on empty coalesce fields

ApplicationGlobal

Created2026-09-28 04:17:06

\* Target tableEmployee Test [u\_employee\_test]

Order100

Run script

CopyUpdateDelete

Related Links

[Auto Map Matching Fields](#)

[Mapping Assist](#)

[Validate Coalesce Fields](#)

[Transform](#)

[Index Coalesce Fields](#)

[Run Point Scan](#)

Field Maps (5)Transform ScriptsUnused coalesce fields for empty values (1)

| Source field    | Target field    | Coalesce |
|-----------------|-----------------|----------|
| u_email         | u_email         | false    |
| u_city          | u_location      | false    |
| u_account_type  | u_department    | false    |
| u_customer_id   | u_employee_id   | true     |
| u_customer_name | u_employee_name | false    |

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AllFavoritesHistoryWorkspacesAdmin

Import Set - ISET0010003

Search

UpdateDelete

Import Set ISET0010003

Related Links

Reprocess

Import Set Runs (1)Import Set Rows (22)Import Log

CreatedSearch

Set = ISET0010003

| Created             | Row | State   | Target record                              | Error   | Comment                 | Transform Map                         |
|---------------------|-----|---------|--------------------------------------------|---------|-------------------------|---------------------------------------|
| 2026-09-28 07:23:24 | 0   | Ignored | Employee Test: Created 2026-09-28 04:23:25 | (empty) | No field values changed | Employee Staging to Employee Test Map |
| 2026-09-28 07:23:24 | 1   | Ignored | Employee Test: Created 2026-09-28 04:23:25 | (empty) | No field values changed | Employee Staging to Employee Test Map |
| 2026-09-28 07:23:24 | 2   | Ignored | Employee Test: Created 2026-09-28 04:23:25 | (empty) | No field values changed | Employee Staging to Employee Test Map |
| 2026-09-28 07:23:24 | 3   | Updated | Employee Test: Created 2026-09-28 04:23:25 | (empty) |                         | Employee Staging to Employee Test Map |
| 2026-09-28 07:23:24 | 4   | Ignored | Employee Test: Created 2026-09-28 04:23:25 | (empty) | No field values changed | Employee Staging to Employee Test Map |
| 2026-09-28 07:23:24 | 5   | Ignored | Employee Test: Created 2026-09-28 04:23:25 | (empty) | No field values changed | Employee Staging to Employee Test Map |
| 2026-09-28 07:23:24 | 6   | Ignored | Employee Test: Created 2026-09-28 04:23:25 | (empty) | No field values changed | Employee Staging to Employee Test Map |
| 2026-09-28 07:23:24 | 7   | Ignored | Employee Test: Created 2026-09-28 04:23:25 | (empty) | No field values changed | Employee Staging to Employee Test Map |
| 2026-09-28 07:23:24 | 8   | Ignored | Employee Test: Created 2026-09-28 04:23:25 | (empty) | No field values changed | Employee Staging to Employee Test Map |
| 2026-09-28 07:23:24 | 9   | Updated | Employee Test: Created 2026-09-28 04:23:25 | (empty) |                         | Employee Staging to Employee Test Map |
| 2026-09-28 07:23:24 | 10  | Ignored | Employee Test: Created 2026-09-28 04:23:25 | (empty) | No field values changed | Employee Staging to Employee Test Map |
| 2026-09-28 07:23:24 | 11  | Ignored | Employee Test: Created 2026-09-28 04:23:25 | (empty) | No field values changed | Employee Staging to Employee Test Map |
| 2026-09-28 07:23:24 | 12  | Ignored | Employee Test: Created 2026-09-28 04:23:25 | (empty) | No field values changed | Employee Staging to Employee Test Map |
| 2026-09-28 07:23:24 | 13  | Ignored | Employee Test: Created 2026-09-28 04:23:25 | (empty) | No field values changed | Employee Staging to Employee Test Map |
| 2026-09-28 07:23:24 | 14  | Ignored | Employee Test: Created 2026-09-28 04:23:25 | (empty) | No field values changed | Employee Staging to Employee Test Map |
| 2026-09-28 07:23:24 | 15  | Ignored | Employee Test: Created 2026-09-28 04:23:25 | (empty) | No field values changed | Employee Staging to Employee Test Map |
| 2026-09-28 07:23:24 | 16  | Ignored | Employee Test: Created 2026-09-28 04:23:25 | (empty) | No field values changed | Employee Staging to Employee Test Map |
| 2026-09-28 07:23:24 | 17  | Ignored | Employee Test: Created 2026-09-28 04:23:25 | (empty) | No field values changed | Employee Staging to Employee Test Map |
| 2026-09-28 07:23:24 | 18  | Ignored | Employee Test: Created 2026-09-28 04:23:25 | (empty) | No field values changed | Employee Staging to Employee Test Map |
| 2026-09-28 07:23:24 | 19  | Ignored | Employee Test: Created 2026-09-28 04:23:25 | (empty) | No field values changed | Employee Staging to Employee Test Map |

servicenow

AllFavoritesHistoryWorkspacesAdmin

Create a report

Search

SaveRun

Edit report

DataTypeConfigureStyle

Report Title: Employees by Department

GeneralTitleLegend

Chart color: Use color palette

Set palette: Default US14

Display data labels

Custom chart size

Chart size: Large

Drilldown view

Decimal precision: 2

Table: Employee Test [u\_employee\_test]

All

Employees by Department

ServiceNow • 7,411,18%Cloud Computing • 5,129,43%Salesforce • 5,129,41%

BackShare

26°C

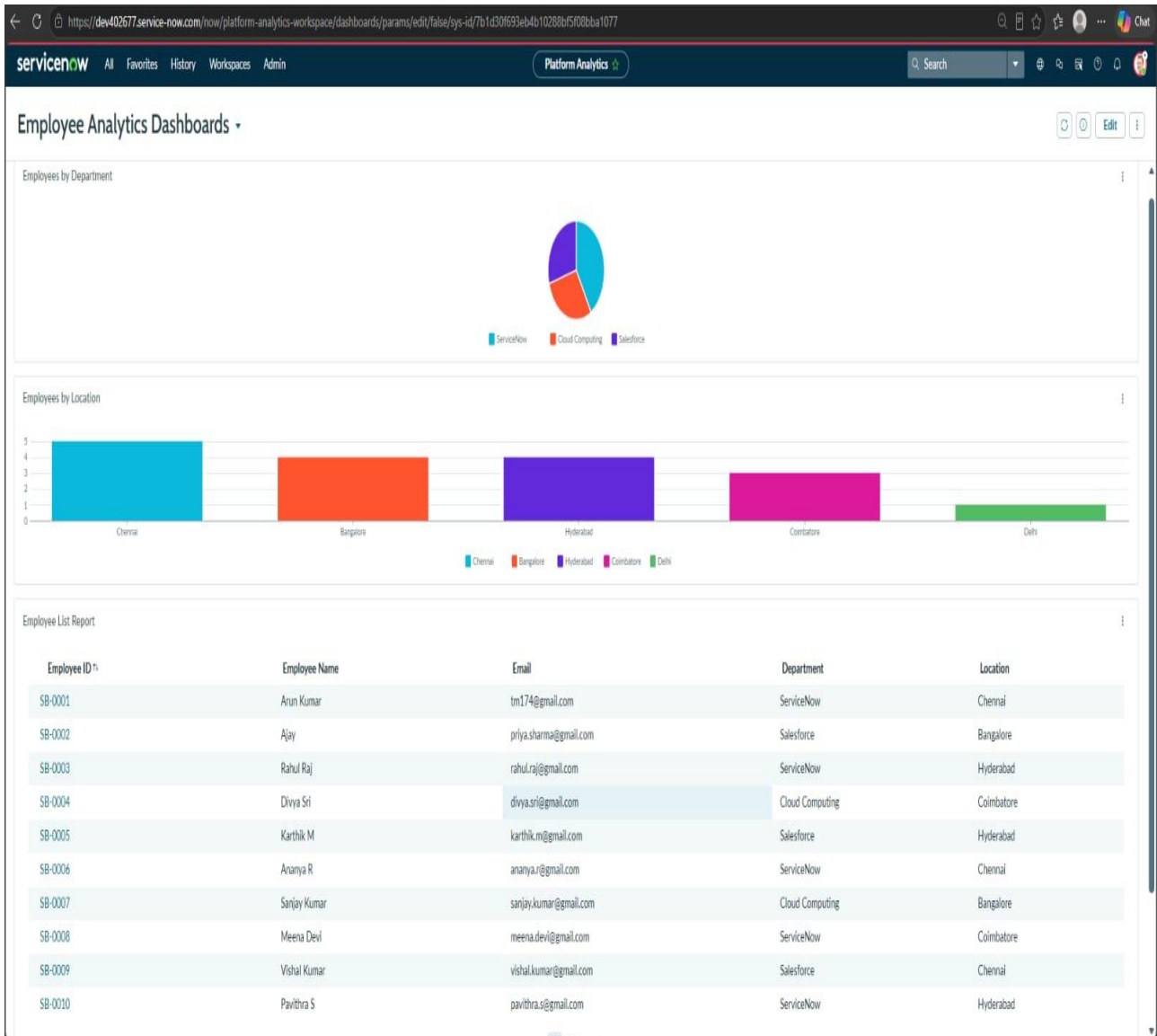
Mostly sunny

Search

ENG IN

10:23

01-10-2026



## 8. ADVANTAGES & DISADVANTAGES

### Advantages

- Reduces manual data entry for bulk employee records.
- Provides structured source-to-target field mapping.
- Supports repeated imports and record updates.
- Coalesce helps prevent duplicate records.
- Transform history provides import validation information.
- Reports and dashboards provide centralized visibility into employee data.

### Disadvantages

- The spreadsheet must follow the expected structure and field format.
- Incorrect field mapping can produce inaccurate target records.
- The project depends on appropriate ServiceNow access and configuration permissions.
- Large or inconsistent source files may require additional data validation before import.

## 9. CONCLUSION

The project successfully demonstrates employee data migration into ServiceNow using Import Sets and Transform Maps. The workflow covers spreadsheet preparation, staging, source-to-target field mapping, transformation and validation. Coalesce extends the solution by identifying matching employee records and supporting updates instead of unnecessary duplicate insertion. Reports and the Employee Analytics Dashboard provide a centralized way to view employee information and import-related results. Overall, the project provides a practical understanding of an automated ServiceNow data import workflow.

## 10. FUTURE SCOPE

- Add additional employee fields and validation rules.
- Introduce automated scheduled imports for regularly received spreadsheets.
- Add more detailed error reporting and data-quality checks.
- Extend dashboards with additional employee and import-performance metrics.
- Integrate the workflow with other external data sources where required.
- Add role-based access controls for different administrative users.

## 11. APPENDIX

### Source Code (if any)

No separate application source code is required for the core workflow; the implementation is configured inside ServiceNow using tables, Import Sets, Transform Maps, field mappings, Coalesce, Reports and Dashboards.

### Dataset Link

<https://docs.google.com/spreadsheets/d/1ci0BReRAHscecwd3dYWmLrDP1lcZRdBT/edit?usp=sharing&ouid=110449630628915607184&rtpof=true&sd=true>

### GitHub & Project Demo Link

<https://github.com/rajikalairaji1-prog/service-now-transform-map-project.git>

<https://drive.google.com/file/d/1bYacbJKLNirJWDXSezjlbTLQlhgUXU66/view?usp=sharing>